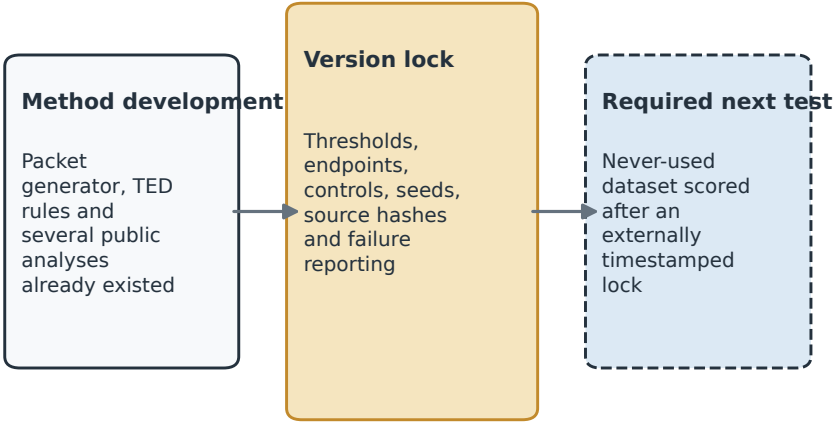


Leakage-audited benchmark design

A. Retrospective lock and prospective-test gap



Current evidence is retrospective; a true post-freeze dataset is not yet available

B. Development, tuning and shifted audit

Partition	n	Replicates	Truth used for selection	Post-freeze
development	80	1-8	yes	no
tuning	40	9-12	yes	no
shifted audit	40	13-16	no	no

Audit outcomes were masked during prediction, but the generator and TED rules pre-date the split.

C. Evidence strata and inferential units

Stratum	Truth/readout	Unit
Controlled packets	latent event/artifact truth	packet
ZSCAPE holdout	full-fit event object	embryo
GSE153056	PD-L1 protein readout	replicate block
SCP1064	RNA-protein + shuffles	block/guide/target
GSE271399	3-day directional sensitivity	condition/day matrix; no independent replicate

D. Current-task comparators and leakage audit

No baseline or TED output is treated as post-freeze external evidence

Executable packet baselines
Most-frequent dummy, multinomial logistic regression, random forest and histogram gradient boosting

Selection rule
Hyperparameters and strongest comparator fixed from tuning results before shifted-audit scoring

Native upstream adapters
tradeSeq, GSVA, AUCell and POT retain native-task evaluation; absent TED fields are coverage gaps